

On Connectome and Geometric Eigenmodes of Brain Activity

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On Connectome and Geometric Eigenmodes of Brain Activity: The Eigenbasis of the Mind?

Measuring what happens inside the human brain is an open and complex problem. Despite the development of many different neuroimaging techniques, it is still unclear how to measure brain dynamics that are directly relevant to phenomenology. However, in recent years, promising mathematical frameworks that explain brain activity and plausibly relate with phenomenology have emerged.

First, in 2016, Selene Atasoy and colleagues developed what's known as the "connectome-specific harmonic waves" (CSHW) framework. CSHW introduced a new frequency-specific representation of cortical activity based on the functional networks of the human brain. Later, in 2023, James Pang and colleagues derived these frequency-specific representations from the shape of the brain, which they called geometric eigenmodes. They then demonstrated how geometric eigenmodes yielded greater accuracy in explaining brain activity while providing a simpler model.

But what are these frequency-specific ways of measuring brain activity? Why is it better to derive them from the shape of the brain? Why are they plausibly directly related to phenomenology? And why is QRI excited about them? Let's explore these questions together.

The eigenbasis of brain dynamics

In physics and engineering, eigenmodes are particular modes in which a system vibrates. They are distinct patterns of motion in which a system resonates. They are crucial in understanding the behavior of complex systems. Mathematically speaking, eigenmodes relate to special vectors which preserve their direction after a linear transformation. Methods from linear algebra determine the frequencies (eigenvalues) and the patterns of motion (eigenvectors) associated with the eigenmodes of a given system.

The concept of eigenmodes extends to the brain. Brain or neural eigenmodes are patterns of synchronized activity that arise from the collective behavior of interconnected neurons in the brain. Eigenmodes are orthogonal, so they are a basis set that can be used to decompose spatiotemporal dynamics as a weighted sum. However, they can be derived in different ways, leading to capturing different patterns.

The CSHW framework derives eigenmodes from the connectome, an undirected and unweighted graph representation of the human brain. More specifically, the CSHW connectome is constructed by sampling ~20k nodes from cortical gray matter, constructing short-range edges from the cortical surface mesh of MRI data, and long-range connections from white-matter cortico-cortical and thalamo-cortical fibers in DTI data. From the connectome graph, we can obtain an adjacency and a degree matrix which provide the graph Laplacian. Solving the eigenvalue problem over the graph Laplacian gives the connectome eigenmodes.

Instead, geometric eigenmodes come from the shape of the brain. Pang and his co-authors considered the cerebral cortex as a 2D model embedded within 3D Euclidean space and used a mesh representation of a population-averaged template capturing local vertex-to-vertex spatial relations and curvature. Then they constructed the Laplace-Beltrami operator (LBO) from this surface mesh and solved the eigenvalue problem.

Importantly, when the number of nodes in a graph increases, the graph Laplacian converges to the Laplace-Beltrami operator. Thus, connectome and geometric eigenmodes are mathematically equivalent and what differs is the underlying data (connections and shape) used by either framework.

Connectome eigenmodes are distinct patterns of activity determined by the interactions between interconnected and functionally specialized cell populations. Intuitively, they may relate to concepts such as how to efficiently partition the

brain connectome or the rate of convergence of a signal traveling randomly across the brain network. Differently, geometric eigenmodes are determined by fundamental aspects of the neocortex geometry such as overall size and shape, curvature and foldings of the gyri.

Eigenmodes explain brain activity

Now that we know how to construct different eigenbasis of brain dynamics, the next question to ask is how do we know that these really represent what happens in the brain? And given that we described two different ways of deriving them; which one better describes brain activity?

In their 2016 study, Atasoy and her colleagues demonstrated the relation between connectome eigenmodes and brain activity by calculating the mutual information and F-measure (a combination of precision and recall of prediction) between connectome eigenmodes and resting state networks in the brain (low-frequency activity strongly temporally correlated across the brain). What they found was correlation between the visual, somato-motor and limbic networks with low-frequency eigenmodes, and correlation between control, dorsal attention and ventral attention with a broader range of connectome eigenmodes. Notably, they found a statistically significant similarity between the default mode network (DMN) and an eigenmode in the range of the 9th connectome eigenvalue for all human subjects analyzed ($n = 10$). In a subsequent 2021 study, Atasoy and colleagues scaled their evaluation to 812 fMRI images from the Human Connectome Project (HCP) finding even clearer evidence of the overlap between patterns of activations of the connectome eigenmodes and functional regions.

Pang et al. (2023) used brain activity of 255 healthy individuals from the HCP to evaluate how accurately geometric eigenmodes capture task-evoked and spontaneous brain activity. They parcellated brain regions and quantified accuracy by computing the correlation between empirical and reconstructed task-evoked activation maps for task-evoked data and spontaneous functional coupling matrices for resting state data. Finally, they showed that with data parcellated over 180 brain regions, the first 10 geometric eigenmodes reconstructed brain activity correlated at $r = 0.38$ with empirical data and the first 100 eigenmodes achieved $r > 0.8$ reconstruction accuracy.

Furthermore, Pang and his colleagues also compared the reconstruction accuracy of geometric eigenmodes with other types of brain eigenmodes including: connectome eigenmodes, eigenmodes from a synthetic connectome constructed with a stochastic wiring process governed by an exponential distance-dependent connection probability (EDR connectome) and a third set derived from the empirical connectome thresholded to match the density of the EDR connectome. Geometric eigenmodes consistently showed the highest reconstruction accuracy across both task-evoked and spontaneous brain activity, with connectome eigenmodes being the least accurate of the four methods. However, specifically for task-evoked data, there was a slight performance advantage for connectome eigenmodes for reconstructions incorporating fewer than ten modes.

Among additional empirical evaluations of geometric eigenmodes, a result that is especially worthy of mention emerged from the application of geometric eigenmodes to non-cortical structures. Pang and colleagues were able to apply their framework on the hippocampus, thalamus, and striatum by calculating the geometric eigenmodes with a tetrahedral mesh. Strikingly, the spatial profiles of the first three functional gradients of the subcortical structures had a near-perfect match with the first three geometric eigenmodes with a mean reconstruction accuracy above 0.92.

Which framework is better?

Among frameworks attempting to measure brain dynamics, geometric eigenmodes are an exciting development. When compared with connectome eigenmodes, they provide a simpler and more mathematically elegant framework that is also easier to apply in practice, since it only requires anatomical images and a mesh representation. On the contrary, connectome eigenmodes require MRI and DTI images, a graph-based model of macroscopic interregional connectivity, the definition of graph nodes and the application of a thresholding procedure to remove putatively spurious connections. Furthermore, Pang's experiments demonstrated that geometric eigenmodes can reconstruct empiric brain activity more accurately.

Additionally, while connectome eigenmodes use a different connectome graph for each subject, geometric eigenmodes use a mesh representation of a population-averaged template of the neocortical surface, fitted to each subject, task and

time frame using an amplitude parameter. Thus, geometric eigenmodes appear to be more invariant across people, providing a potentially more generalisable framework. Finally, in relying on geometry rather than connectivity, geometric eigenmodes treat the brain as a continuous system rather than functionally discrete one which is an idealized conceptualisation. Notably, these results are consistent with the general intuition that geometry poses a more fundamental constraint on the behavior of a system. For example, in fields such as cymatics, standing waves are determined only by two factors: the geometry of the metal plate and the frequency of vibration.

However, it is important to note that geometric eigenmodes were constructed using unihemispheric data. The authors claim that their approach can be easily extended to the whole brain but it would be important to confirm the results across both hemispheres given known evidence of functional and geometric differences between the two hemispheres. Also, it is speculatively possible that the higher generalization ability of geometric eigenmodes may come at the expense of explaining finer differences between two brains (even of the same person across time or conditions) that don't involve changes in shape.

Eigenmodes model brain activity over time

Both connectome and geometric eigenmodes assume that information travels through the brain as waves. Thus, both of them can be integrated in neural field models that predict brain activity over time. This extra step in the application of brain eigenbasis helps determine which mechanism lies behind the self-organization of oscillatory activity.

Atasoy et. al (2016) identified several biological mechanisms (such as somatostatin-expressing inhibitory neurons and slow synaptic transmission caused by N-methyl-D-aspartate receptors contributing to excitatory currents) that when taken together seem to give rise to functional circuitry equivalent to short-range excitation coupled with broad inhibition. This type of functional circuitry, known as the Mexican hat organization, is the necessary condition for self-organization in neural field models based on the Wilson-Cowan differential equations. The authors incorporated the connectome Laplacian into the diffusion term of the Wilson-Cowan equations, thus extending a variant of the neural field model based on these equations to the 3D connectome model. The authors observed that for a wide range of diffusion parameters, oscillatory patterns self-organize, and through linear stability analysis, they could activate a wide range of connectome harmonics, making the neural field model a plausible mechanism for the self-organization of connectome eigenmodes.

Going beyond this, Pang et. al (2023) modeled neural activity over time using a simple neural field theory wave equation whose spatial part satisfies the Helmholtz equation (which is equivalent to the Laplacian-Beltrami Operator of the neocortical mesh representation). They then compared the behavior of this model with a balanced excitation-inhibition model aligned with the connectome-centric view of brain function. Overall, the wave model showed comparable or superior performance in reconstruction of empirical data relative to the neural mass model. Strikingly, when the wave model was given an 1ms input to the primary visual area (V1) of the cerebral cortex, it yielded a propagating wave that splits along the dorsal and ventral visual processing streams consistent with the mainstream understanding of hierarchical visual processing. An impressive result!

Is the brain a non-linear wave computer?

The fact that neural eigenmodes can accurately reconstruct brain activations in space and reproduce brain activity over time when embedded into wave models is strong evidence to support QRI's nonlinear wave computing paradigm. This is the idea that the brain uses combinations of linear and nonlinear waves (i.e waves with behavior more complex than superposition) to represent and compute things. If nonlinear wave computing is indeed what gives rise to consciousness, then brain eigenmodes capture how such a system is implemented and can measure its behavior.

The idea that waves computing and nonlinearities relate to how the brain and consciousness function was originally put forth by Steven Lehar. In his attempt to try and explain how visual perception constructs objects that we consider real (Lehar, 2008) he noted that visual perception generates a complete virtual world of experience based on a relatively poor sensory input. He observed that the brain is probably using processed raw 2D images from the eyes to construct every possible spatial interpretation in parallel and then selecting the subset of patterns that embody the greatest intrinsic symmetry. Much like an echo chamber that picks out its resonant frequency from a white noise stimulus containing all frequencies. Such an algorithm, which he called the forward-and-reverse grassfire algorithm, exhibits many of the

properties of a nonlinear standing wave model. A great starting point to dive deeper into the fascinating work of Steven Lehar and non-linear wave computing is Cube Flipper and Scry's introduction to his work .

Linking eigenmodes, nonlinear waves and phenomenology

If the brain indeed self-organizes around eigenmodes and constructs objects using non-linear waves, as evidence seems to be pointing at, then these provide a powerful way to efficiently compress a lot of the complexity of the brain. As pointed out by Johnson (2017) “we have accumulated a lot of experimental knowledge about how neurons work and observational knowledge about how people behave, but we have a few elegant compressions for how to connect the two”. Brain eigenmodes, and geometric eigenmodes in particular, appear to be a powerful and parsimonious way of linking measurable brain activity with psychological phenomena. Being able to directly measure the underlying neurological manifestation of experiences can give us much more understanding about and control over them.

For example, in a 2017 study, Atasoy and other researchers studied the connectome eigenmodes of 12 healthy participants who received 75µg of LSD and analyzed the correlation between experience and the energy of an eigenmode (ie the combination of the strength of activation of a particular eigenmode and its own intrinsic energy). They found a significant correlation between changes in the energy of low frequency eigenmodes and ego dissolution experiences and emotional arousal, and between positive moods and changes in high frequency eigenmodes. They also found that LSD increased the probability of activating unusual brain states. This happened non-randomly by increasing the co-occurrence probability of different eigenmodes. This suggests that LSD facilitates the emergence of unusual brain states as the combination of eigenmodes that tend to not co-occur during normal consciousness, which is consistent with the phenomenology of LSD experiences.

Going further, building upon the idea of the brain as a nonlinear wave computer, supported by the empirical validity of neural eigenmodes, Andrés Gomez Emilsson suggested that the shape of the waves that are traveling through the nervous system and how they travel may explain the qualities of the experiences that they generate, such as whether they are pleasant or not, as suggested by valence structuralism (Johnson, 2016). This closely aligns to what we commonly call the vibe of an experience. For example, pure MDMA generally has incredibly soft vibes which may result from a constant dithering of the nervous system which soften all waves. Going further, he suggests that following Lehar's work, the combination of these waves constructs real objects in our world simulation and worldviews based on self-reinforcing objects whose pleasantness is determined by interaction of the underlying waves, or brain eigenmodes in neuroscientific terms.

The power of measuring states of mind

The development of methods to measure and compress complex brain activity is exciting. Their empirical validity and consistency with theories about complex system dynamics strengthens their potential of providing a base for unambiguously linking measurable brain states with experience. This trajectory of development is something that QRI is excited about because it lays the foundation for developing a precise mathematical language for describing subjective experience and building technologies to improve the lives of sentient beings.

Precise eigenbasis of brain dynamics can support numerous impactful applications such as parameterising and measuring mental health (as described by Andrés Gomez Emilsson, 2017, in his presentation about quantifying bliss). In turn, it can enable us to identify a target emotional state for someone and use knowledge about the eigenbasis of the current and target state to inform us how to push the system toward that state.

Johnson (2017) suggests many other exciting avenues that can open as a result of developing an efficient measurement of brain activity that is tightly coupled with phenomenology, such as:

- reconceptualizing intelligence as the capacity to cleanly switch between different complex neuroacoustic profiles;
- reconceptualizing emotional intelligence as the ability to entrain own's brain eigenmodes with those of another person and/or robustly signal out one's internal eigenbasis state so that others be entrained to it;
- a neuroscientific basis for the big 5 personality framework;

- a mathematical framework for understanding the nature and signal propagation of emotions;
- a direct window into language as a felt-sense phenomenon;

Furthermore, frameworks leveraging eigenbasis of brain dynamics, which are still relatively young, need refinement and addressing of open questions. For example, how do geometric eigenmodes change during neurodevelopment? Or as a result of brain injury? Can geometric eigenmodes explain diseases that significantly affect brain activity and phenomenology without affecting shape? Answers to such questions are likely to come in the near future, so stay tuned as the future of neuroscience is going to be real fun.

Tags

neuroscience , consciousness , phenomenology

References

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